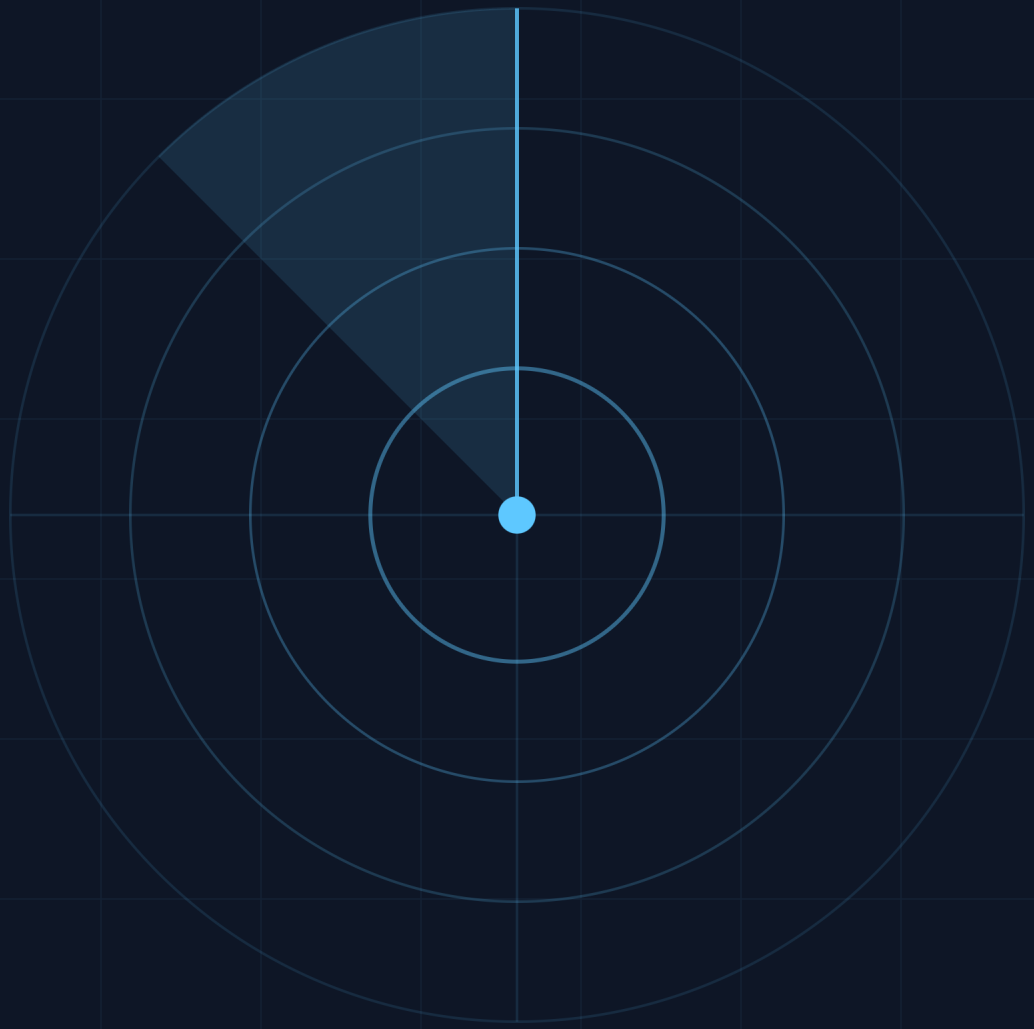
ENGLISH PRESENTATION · SIIT

# Your WiFi Can See You

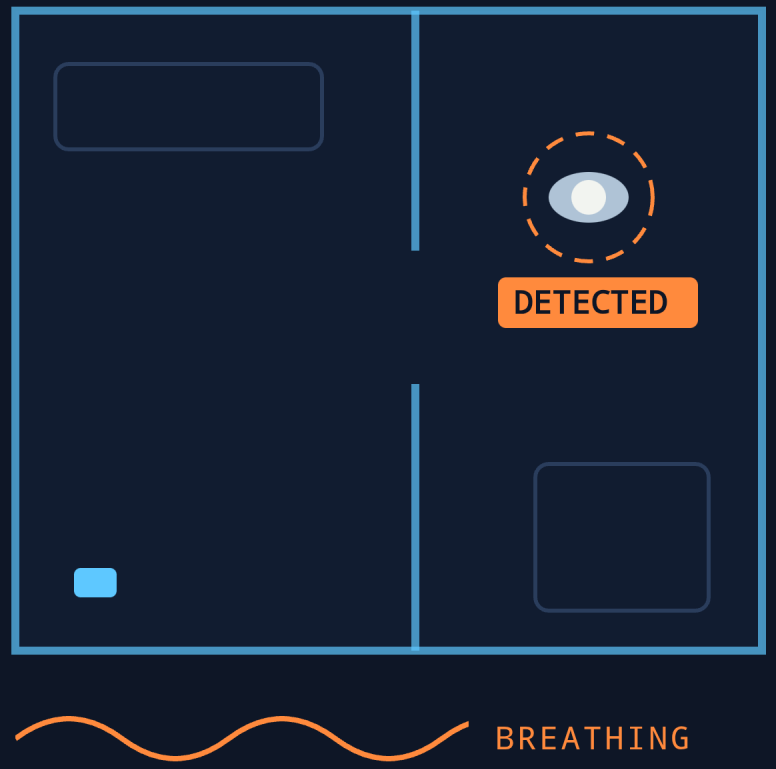
How a normal router can work like a radar

Stefan Petrović



A QUESTION FOR YOU

# What if your router could see you, even through a wall?



# How it works



01 · SEND

## The router sends radio waves

All the time, in every direction.



02 · BOUNCE

## The waves bounce off your body

We are mostly water, and water reflects WiFi.



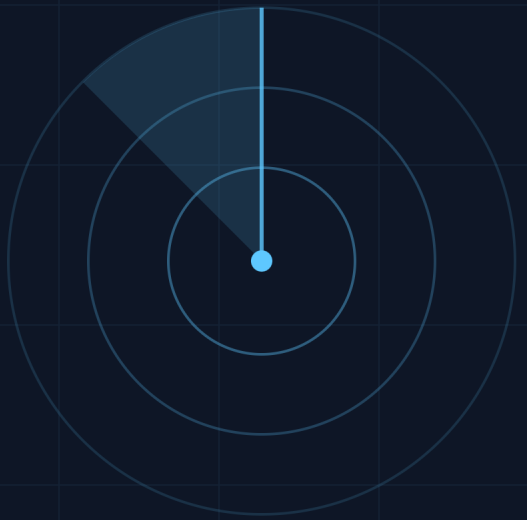
03 · MEASURE

## A computer reads the changes

When you move, the signal changes too.

# From WiFi to radar

A radar sends waves and listens to the echo. With AI, a router can do the same.



MIT · Prof. Dina Katabi

**Seeing movement and breathing through walls**

Carnegie Mellon · 2023

**“DensePose from WiFi”: full body pose with no camera**

# The good side



## Fall detection

Help for older people who live alone.



## Breathing and sleep

Health checks with no wires on the body.

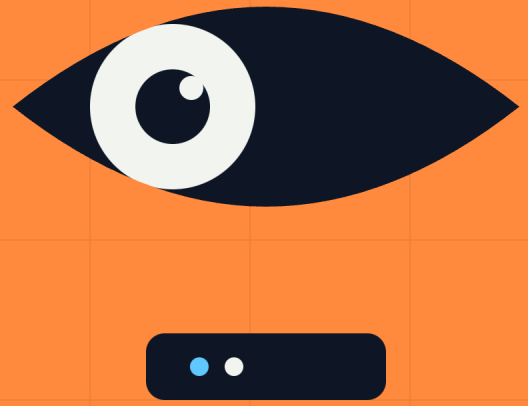


## Smart homes

Lights and heating that know you are home.

## The dark side

# No camera.
# No permission.
# No idea.



Someone could know if you are home, where you are and what you are doing.

TO SUM UP

# WiFi was made to connect us. Now it can also see us.



# Who is looking?

# Thank you!

Any questions?